

Lesson 5: Scoring Visual Text Analytics Models



5.0 Lesson Overview



Lesson Overview

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



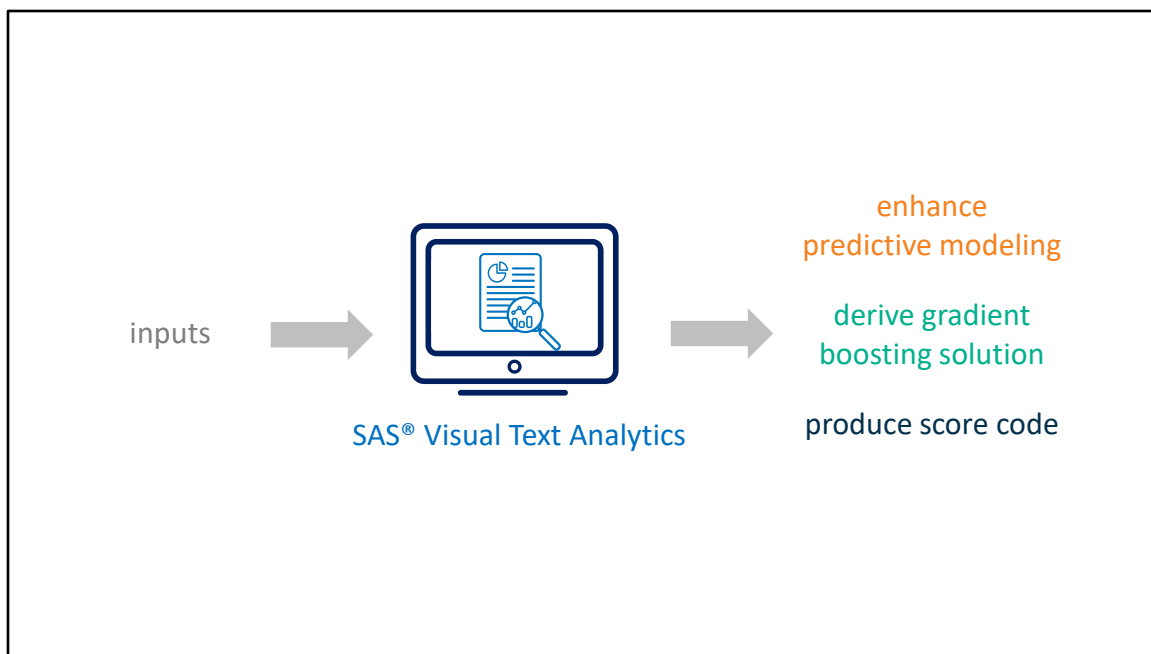
5.1 Introduction to Text Analytics Scoring



Integrating SAS Viya Projects and Software



Copyright © SAS Institute Inc. All rights reserved.



Although SAS Visual Text Analytics provides a powerful software environment for text processing and analysis, there are situations where text analytics is one of several components of a larger project. For example, text analytics inputs can be used to enhance predictive modeling, where additional transactional and operational inputs can be combined to derive a gradient boosting solution supported by SAS Visual Data Mining and Machine Learning.

SAS Visual Text Analytics nodes produce score code that can generate data for analysis using other SAS products.



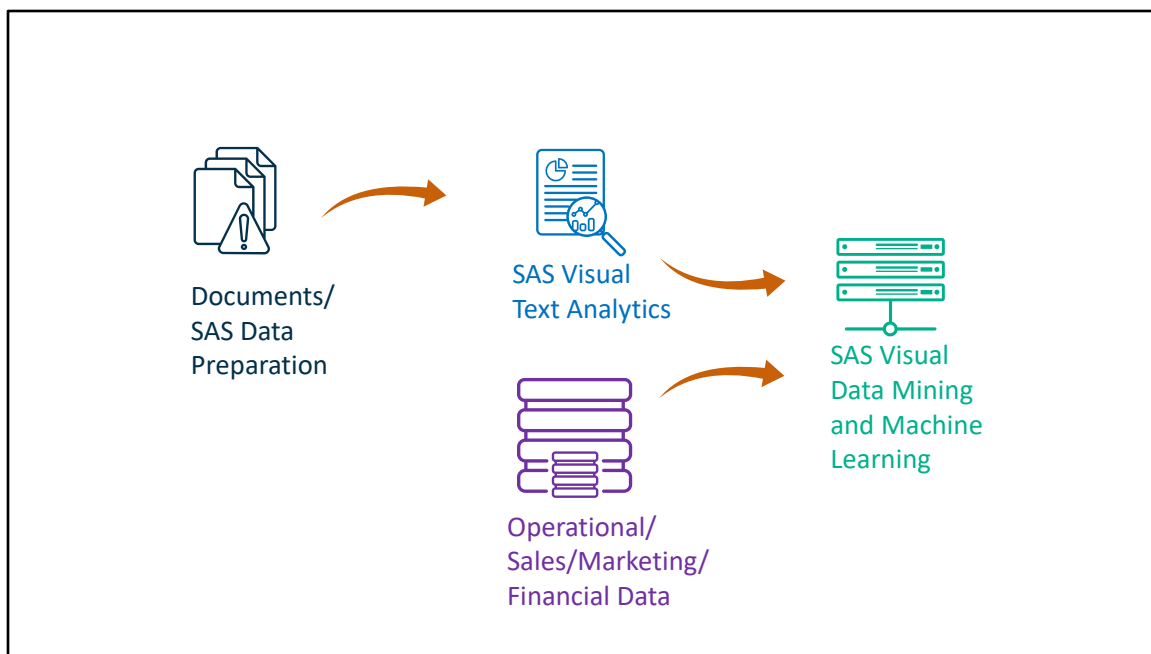
SAS® Visual Text Analytics

- Expand assessment of models by using SAS Visual Analytics to explore scored data.
- Enhance predictive modeling in SAS Data Mining and Machine Learning by using inputs from SAS Visual Text Analytics.
- Add alternative algorithms to categorize documents by exporting SAS Visual Text Analytics results to SAS Visual Data Mining and Machine Learning.

You can expand assessment of SAS Visual Text Analytics models by using SAS Visual Analytics to explore scored data.

You can enhance predictive modeling in SAS Data Mining and Machine Learning by using inputs from SAS Visual Text Analytics.

And you can add alternative algorithms, such as gradient boosting and neural networks, to categorize documents by exporting SAS Visual Text Analytics results to SAS Visual Data Mining and Machine Learning.



Document processing and conversion is supported by SAS Data Preparation, which is accessed through the Explore and Visualize data task in the show list of applications menu.

SAS Visual Text Analytics can derive concepts, topics, and categories for documents. Scored documents produce additional variables derived from the concepts, topics, and categories score code. The scored data can be combined with additional data, such as operational, sales, marketing, and financial data. The combined data can be processed by SAS Visual Data Mining and Machine Learning to produce solutions for supervised and unsupervised learning.

5.2 Scoring Concepts



Overview of the ASRS Project



Copyright © SAS Institute Inc. All rights reserved.

ASRS Data



- Voluntary safety reporting system
 - NASA
 - FAA

In this section, we're going to work with ASRS (Aviation Safety Reporting System) data. The ASRS is a voluntary safety reporting system. It's supported by the U.S. National Aeronautics and Space Administration, with assistance from other U.S. government agencies, like the U.S. Federal Aviation Administration.

ASRS Data



- Identify airports mentioned
- Determine most frequently occurring airports

The ASRS project seeks to identify all airports mentioned in the ASRS safety reports. One deliverable from the project is a list of the most frequently occurring airport references.

Demo: Scoring Airport Concepts, Part 1

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Scoring Airport Concepts, Part 2

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Scoring Airport Concepts, Part 3

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Practice: Extracting Numeric Data from Text (Self-Study)

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



5.3 Scoring Topics



Overview of ASRS Topics



Copyright © SAS Institute Inc. All rights reserved.

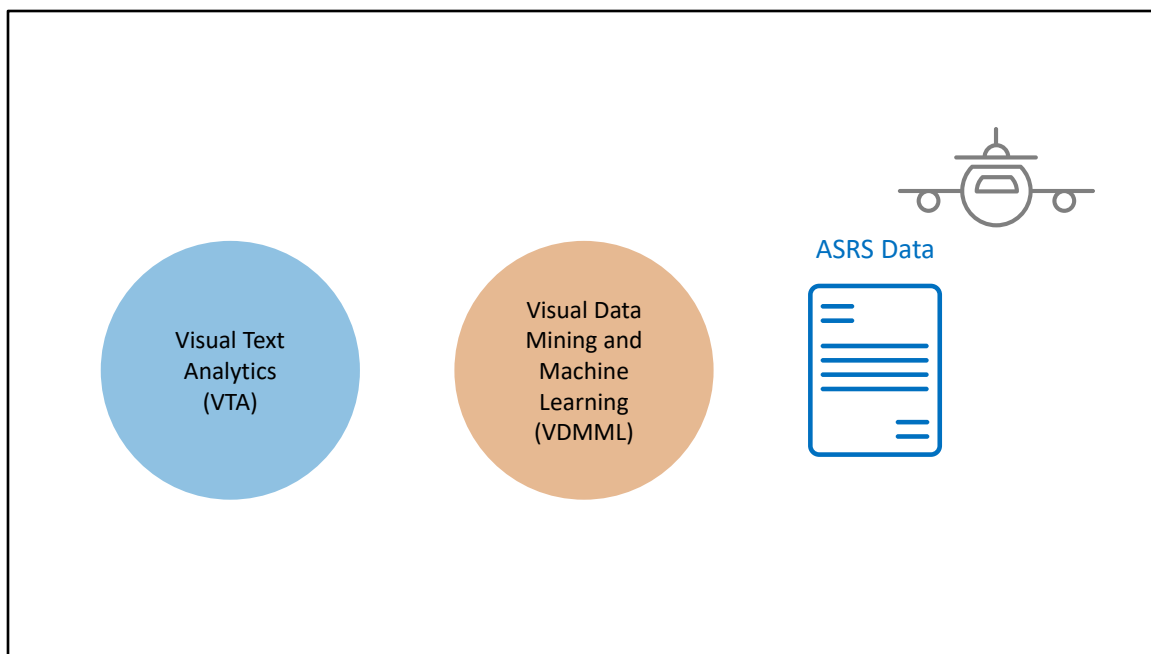
Topics (14)						
☐	Topic	Created by	Documents ↓ ?			
☐	tower, clearance, takeoff, hear, runway	System	3430			
☐	knot, gear, brake, flap, left	System	3126			
☐	flap, takeoff, knot, flight, dispatch	System	3090			
☐	approach, visual, runway, visual approach, +locate	System	3005			
☐	+foot, descend, altitude, +autopilot, cross	System	2991			
☐	engine, emergency, declare, smoke, cabin	System	2856			
☐	install, maintain, inspect, minimumequipmentlist, mechanic	System	2607			
☐	airspace, visualflightrules, classb, area, plan	System	2605			
☐	degree, head, turn, radial, degree radial	System	2515			
☐	trafficalertandcollisionavoidancesystem, resolutionadvisory, traffic, +foot, climb	System	2194			
☐	taxiway, taxiway, taxi, ramp, groundcontrol	System	2078			
☐	passenger, flightattendant, seat, captain, hi	System	2057			
☐	hold, line, short line, short, runway	System	1996			
☐	flightlevel, descend, aircraftnumber, center, +aircraft	System	1982			



ASRS Data



The ASRS safety reports generated 14 computer-derived topics. The Topics node in SAS Visual Text Analytics generates a set of numeric measurements for each document related to a set of computer derived or human generated topics. These numeric measurements can be used as inputs in a predictive model.



SAS Visual Text Analytics provides score code to generate the numeric measurements, and SAS Visual Data Mining and Machine Learning exploits these numeric measurements, using them as inputs for a predictive model that is constructed to accurately categorize the reports with respect to known safety labels.

Demo: Scoring ASRS Topics

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



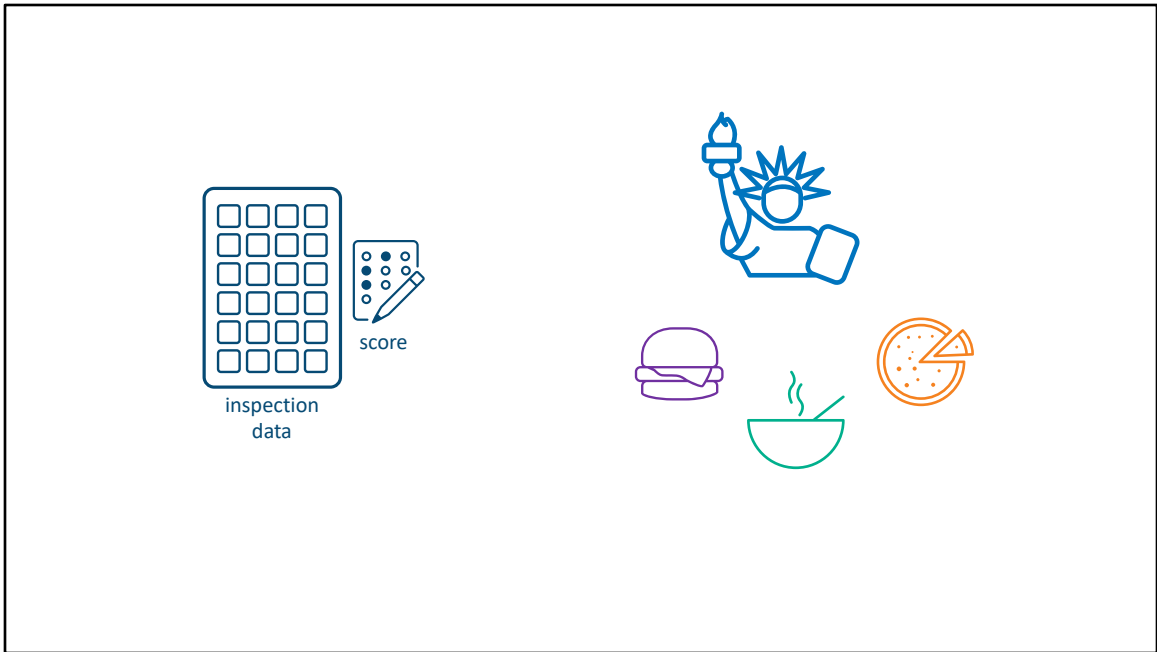
5.4 Scoring Categories



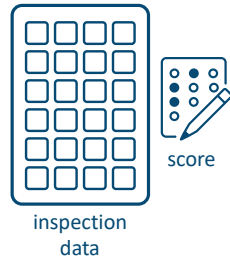
NYC Open Data Restaurant Inspections



Copyright © SAS Institute Inc. All rights reserved.



Now let's score a document collection with respect to categories. We'll use the New York City restaurant inspections data, which are available online as part of New York City Open Data. The data are provided by NYC Department of Health and Mental Hygiene.



- 21,544 restaurants as of January 19, 2020
- 20,530 restaurants in the analysis data set

The inspections were for 21,544 restaurants as of January 19, 2020. After cleaning up the data, only 20,530 restaurants remain in the analysis data set.

Demo: Scoring Categories, Part 1

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Scoring Categories, Part 2

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Scoring Categories



Copyright © SAS Institute Inc. All rights reserved.

Question: Scoring Categories

Which of the following SAS Visual Analytics nodes does *not* produce score code?

- a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

Answer: Scoring Categories

Which of the following SAS Visual Analytics nodes does *not* produce score code?

- a. Concepts
- ☒ b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

b

The Text Parsing node creates the term table that is used by all subsequent nodes, but there is no score code for text parsing. The node serves as a building block for other nodes. All other nodes score documents to extract information or to classify documents with respect to categories or topics.